AAE 352 – Aircraft Structural Analysis I

3 Credits

1. Course Information/Description:

This course introduces aerospace engineering students to properties of wing and fuselage sections, basic concepts in elasticity, torsion of single and multi-cell thin-walled and skin stringer sections, bending and flexural shear in open and closed thin-walled and skin-stringer sections, flexural shear flow in thin-walled sections, and failure criteria for isotropic materials

2. Brightspace Learning Management System:

Course videos, materials, and resources are available through Purdue's Brightspace system; students should begin with the Start Here tab to familiarize themselves with course organization, navigation, and available campus support resources.

3. Textbook:

The primary textbook for this course is Sun, C. T. (2006). Mechanics of Aircraft Structures, 2nd Edition, Hoboken, NJ: Wiley, ISBN 978-0471699668. Additional readings: Lecture notes will be posted to Brightspace after each lecture.

4. Attendance Policy:

Students are expected to attend all scheduled classes in accordance with Purdue University policy. Attendance contributes 2% of the final grade and is measured through 10 unannounced iClicker checks during the semester, with missed checks proportionally reducing the attendance score.

5. Assignments:

To support student learning and evaluate understanding throughout the course, a variety of assignments and assessments will be administered. These are designed to provide ongoing practice, reinforce key concepts, and measure both conceptual knowledge and problem-solving skills. The breakdown of assignments and their contribution to the final grade is as follows:

Assignments and Exams	Due	Points (Traditional Section)
Homework	Weekly	20
Attendance	10 counts	2
Exam 1	Monday 02/23	26
Exam 2	Monday 03/30	26
Exam 3	TBD	26
		Total: 100

6. Extra Credit:

Extra credit of 1% will be given to students who complete mid-term and final course evaluations.

7. Course Learning Objectives:

On completing this course, you shall be able to:

- Describe the material behavior of deformable bodies when subjected to loads.
- Create and use free-body diagrams to compute resultant support forces and moments.
- Compute internal force and moment distributions.
- Properly select and use materials based upon their mechanical properties.
- Compute stresses, strains, deformation, and displacements of basic structural components.
- Predict structural integrity based on the calculations.
- Apply these ideas for analysis of components related to aerospace applications.

8. Evaluation:

Your grade is based on your performance in the homework and exams. Assigned letter grade cutoffs are as follows:

Percentage	Letter grade	Percentage	Letter grade
96.67 – 100	A+	73.33 – 76.66	C
93.33 – 96.66	A	70 – 73.32	C-
90 – 93.32	A-	66.67 – 69.99	D+
86.67 – 89.99	B+	63.33 – 66.66	D
83.33 – 86.66	B	60 – 63.32	D-
80 – 83.32	B-	0 – 59.99	F
76.67 – 79.99	C+		

Please note that final grades are conclusive and will not be changed under any circumstances.

1 AI Policy:

In this course, the use of A.I. is allowed for purposes such as finding or creating additional or new exercises related to the course material. A. I. tools cannot be used for partially or entirely solving or answering homework questions.